

NACS 645 – Preparing the final paper

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From research papers to opinion papers

Tips for writing opinion articles.
Nat Rev Phys **7**, 129 (2025).

“A good opinion article is enjoyable to read, informative, and may even change someone’s mind.”

— *Nature Reviews Physics* (2025)

Type	Core goal	Typical example
Empirical	Present and test new evidence	Research Article, Registered Report
Interpretive / Viewpoint	Argue, reinterpret, or redirect thinking	Perspective, Opinion, Commentary, Comment

Empirical = evidence-driven

Viewpoint = argument-driven

Viewpoint writing

Authors usually consider writing a perspective, or opinion article when:

- Research has been done, but the data are preliminary, not very significant, or not strong enough to convey the author's message.
- The topic is of broad concern to a wider audience or to the scholarly community in general.
- The author wishes to present opinions and ideas or describe an innovation that has not yet been implemented.

Viewpoint writing

- Goal: Shift understanding, not report data
- Argument: one strong, defensible idea
- Evidence: selective and strategic, not exhaustive
- Voice: authorial, confident, and accountable
- Evaluation: clarity, originality, and persuasion – no statistical validity

Type	Core function	Data?	Review?	Typical length	Peer review
Opinion/ Comment	Take a stance; argue for or against a view	No	Light	~2,000–2,500 w	Often
Perspective	Reframe the conceptual landscape	Optional	Yes	~3,000–4,000 w	Yes
Commentary	Respond to a specific publication	No	No	~1,000–1,500 w	Usually invited

Example: Nature Neuroscience / Nature Communications Psychology

Content types

The following article types can be submitted to *Communications Psychology*:

[Article](#)

[Registered Report](#)

[Resource](#)

[Comment](#)

[Review](#)

[Perspective](#)

[Primer](#)

[Matters Arising](#)

Comment

Comment is a very flexible format. Comments may be on policy, science and society or purely scientific issues. The main criteria are that they should be of immediate interest to the readership of the journal and should be written in an accessible, non-technical style. Their length is typically 1-4 pages, up to 1,500 words, although some may be longer. Comments do not normally contain primary research data, although they may present 'sociological' data (funding trends, demographics, bibliographic data, etc). For a detailed breakdown of our formatting requirements for Comment articles, please see our [style and formatting checklist](#).

Comment authors must provide a [competing interests](#) statement before publication.

Comments may be peer reviewed at the editors' discretion.

Perspective

A Perspective is intended to provide a forum for authors to discuss models and ideas from a personal viewpoint. They are more forward looking and/or speculative than Review articles and may take a narrower field of view. They may be opinionated but should remain balanced and are intended to stimulate discussion and new experimental approaches.

Perspectives follow the same formatting guidelines and policies as Reviews. For a detailed breakdown of our formatting requirements for Perspective articles, please see our [style and formatting checklist](#). Perspectives are peer-reviewed and include received/accepted dates. Authors must provide a [competing interests](#) and [author contributions statement](#) before publication.

Matters Arising

Important scientific comments and clarifications on peer-reviewed articles published in *Communications Psychology* may be submitted as Matters Arising. The guidelines for Matters Arising are outlined [here](#).

Editorials

Editorials are written by internal editors, editorial board members or guest editors. They focus on topics that the editorial team feel are of particular current interest to their readership.

Example: PNAS

Unsolicited

Brief Reports describe observations of immediate impact that may hold potential to initiate new avenues of research, provide compelling new data on controversies of broad interest and long-standing questions, or present a concise conceptual advance.

- **All Brief Reports are published immediate open access.**
- Brief Reports are limited to 3 pages, which is approximately 1,600 words (including the manuscript text, title page, abstract, and figure legends), and 15 references.
- Supporting information (SI) is limited to extended methods, essential supporting datasets, and videos (no additional tables or figures).
- All Brief Reports follow the [Direct Submission](#) mode of review and are not eligible as member-contributed submissions

Letters to the Editor provide brief comments that allow readers to constructively address a difference of opinion with authors of a PNAS article.

- Letters must be submitted within six months of the first online publication date of the subject article.
- Letters are limited to 500 words, 2 graphical elements (figures or tables), and 10 references. Legends should only include brief descriptions of the figures. Supporting information is not allowed. Data and code used to support the Letter may be posted in a repository and cited in the Letter. Additional discussion, tables, or figures are not allowed.
- Letters may not include references to submitted papers or unpublished results, requests to cite the Letter writer's work, accusations of misconduct, calls for retraction, or personal comments to an author.
- Letters typically publish alongside an Author Response from the authors of the subject article.
- PNAS will consider Letters for Brief Report, Contributed, Direct Submission, Inaugural, Opinion, and Perspective articles.

Opinions are succinct essays that appear in the Front Matter section. Authored by researchers, Opinion pieces further the discourse on a timely topic, or offer a call to action, via a clearly articulated argument that includes novel ideas or proposals. Opinions are not summaries or synopses; the essay's argument should be explicitly stated early on, and then supported in the text. Submissions should also address relevant counter-arguments. All authors are welcome to submit proposals for consideration. We especially welcome a broad range of voices and contributors. Submissions deemed appropriate for the section are reviewed by at least one NAS member, or other expert, who has the relevant expertise. Submissions should be approximately 1,500–2,000 words, not including references and do not require abstracts.

By invitation only

Commentaries call attention to articles of particular note.

Inaugural Articles may be submitted by newly elected NAS members as a way to introduce themselves to the readers of PNAS and may present new ideas or hypotheses or describe the historical development of the member's field. [View the Inaugural Articles collection.](#)

Front Matter is a magazine section that tells the stories of science. Sections include News Features, Science and Culture, Inner Workings, Core Concepts, QnAs, Profiles, and the *Science Sessions* podcast. All of these elements are written or produced by science writers and journalists who receive assignments from PNAS editors.

Perspectives should identify a critical science problem, provide a state-of-the-art assessment, and offer new insights or a new approach to its resolution. The narrative style may vary, but each Perspective should focus on an important area of research and be accessible to a broad scientific audience. Perspectives may help contextualize findings within a field or add a new dimension to previously published research. Current advances and novel insights are encouraged. To submit a proposed article, select "Presubmission for Perspective." Invited Perspectives are reviewed as described in the [Peer Review Process](#). PNAS allows up to a maximum of 10 pages for these articles.

Title

Purpose: Capture interest, signal stance, and promise value

Guidelines:

- Be **informative yet provocative** ($\leq$ 100 characters)
- Indicate the argument or central idea
- Avoid ambiguity or rhetorical titles that mislead
- If needed, pair with a **standfirst**: 1–2 sentences summarizing why it matters

Examples:

- [Using computational models of learning to advance cognitive behavioral therapy](#)
- [A manifesto for a globally diverse, equitable, and inclusive open science](#)
- [How citizens' experience of democracy can actually pave the way to democratic backsliding](#)

Abstract

Purpose: Orient the reader and make them want to read on

Key elements:

- **Topic significance:** one first sentence situating the debate
- **Core argument:** what position you take
- **Implication:** why it matters for research or society
- ~150 words, written **in plain English**, not telegraphic

Structure: An opinion paper's abstract is not a mini-IMRaD (Intro, Methods, Results, Discussion), it's a **mini-argument**

Introduction

Purpose: Establish context and the question in the first paragraph

Guidelines:

- Open with a specific, concrete problem, not a grand statement
- State why now: the timeliness or relevance hook
- Define your scope and audience (who should care)
- End with your thesis sentence (the claim your piece will defend)

Example opening:

The current political climate, particularly in the US, is undeniably shaking the foundations of our scientific and academic institutions; the future is shrouded in uncertainty. However, one thing is clear: change is here.

Understandably, the scientific community has responded defensively, but this moment also demands reflection. We must ask ourselves how we have contributed to widening the gap between scientific purpose and public perception. Not only has public support for science declined, but so has enthusiasm. More concerning is that polling data highlight how trust in science has become a contentious, partisan issue. How have our own actions — and inactions — contributed to this divide? The answers may not change the immediate challenges that science is facing today, but reckoning with them now can help to prevent lasting damage.

Body

Purpose: Develop the reasoning logically, one idea per paragraph

Guidelines:

- 2–4 sections that each **advance one step** in your argument
- Start each paragraph with a **topic sentence** that signals movement
- Support arguments with **selective evidence or examples**, not exhaustive literature
- Use **transitions** (“Yet this assumption overlooks...”, “A more productive view is...”) to build momentum toward the crux of the argument
- Avoid method or results detail; focus on interpretation and consequences

Structure cue:

Problem → Diagnosis → Proposal → Implications

Conclusion

Purpose: Leave the reader with conviction and direction

Guidelines:

- **Synthesize**, don't repeat
- End with the **implication or action**: what should change (e.g., practice, theory, policy)
- Optionally mention **unanswered questions** or **next steps**
- Avoid grand finales ("This will revolutionize X"). For persuasion purposes, precision and clarity

Example closing:

To realize its full potential to fundamentally change the lives of countless people and families in our society — now and for future generations — not only must we reverse the reductions to funding outlined above, but it is imperative that funding for the BRAIN Initiative be increased above its peak levels. From the current precipice, the potential of a new frontier lies ahead, where we could bridge the gap between what is currently known about the brain and the development of broad-ranging treatments for brain diseases and disorders. Rather than squandering this investment and progress, let us take full advantage of what we have already learned and champion BRAIN to lead a renaissance of discoveries that will unlock the many remaining mysteries about our most complex organ.

Caveat:
Formats, lengths and constraints
widely vary from journal to journal

More resources

- Tips for writing opinion articles. *Nat Rev Phys* **7**, 129 (2025). <https://doi.org/10.1038/s42254-025-00815-7>
- Narula, R. (2024). From the editor: On writing a perspectives article—what they are, what they are not (and what they should be). *Journal of International Business Policy*, 7(3), 253-259.
- How to Write a Perspective and Opinion Article. <https://jle.hse.ru/pop>
- Nature Neuroscience. <https://www.nature.com/neuro/content>
- PNAS. <https://www.pnas.org/author-center/submitting-your-manuscript#article-types>

Opinion or Review?

It's a fuzzy game

Discussed opinion papers

- Melnikoff, D. E., & Bargh, J. A. (2018). The mythical number two. *Trends in cognitive sciences*, 22(4), 280-293.
- Spelke, E. S. (1998). Nativism, empiricism, and the origins of knowledge. *Infant Behavior and Development*, 21(2), 181-200.
- Poldrack, R. A. (2006). Can cognitive processes be inferred from neuroimaging data?. *Trends in cognitive sciences*, 10(2), 59-63.
- Jolly, E., & Chang, L. J. (2019). The flatland fallacy: Moving beyond low-dimensional thinking. *Topics in cognitive science*, 11(2), 433-454.
- Krakauer, J. W., Ghazanfar, A. A., Gomez-Marin, A., MacIver, M. A., & Poeppel, D. (2017). Neuroscience needs behavior: correcting a reductionist bias. *Neuron*, 93(3), 480-490.
- Fitch, W. T. (2011). The evolution of syntax: an exaptationist perspective. *Frontiers in evolutionary neuroscience*, 3, 9.

To discuss opinion papers

- Fehr, E., & Gächter, S. (2000). Fairness and retaliation: The economics of reciprocity. *Journal of economic perspectives*, 14(3), 159-182.
- Hardin, G. (1968). The tragedy of the commons. *Science*, 162(3859), 1243–1248.
- Koster-Hale, J., & Saxe, R. (2013). Theory of mind: a neural prediction problem. *Neuron*, 79(5), 836-848.
- Wheatley, T., Thornton, M. A., Stolk, A., & Chang, L. J. (2024). The emerging science of interacting minds. *Perspectives on Psychological Science*, 19(2), 355-373.
- Brette, R. (2019). Is coding a relevant metaphor for the brain?. *Behavioral and Brain Sciences*, 42, e215.
- Collins, A. G., & Cockburn, J. (2020). Beyond dichotomies in reinforcement learning. *Nature Reviews Neuroscience*, 21(10), 576-586.
- Kay, K. N. (2018). Principles for models of neural information processing. *NeuroImage*, 180, 101-109.
- Perconti, P., & Plebe, A. (2020). Deep learning and cognitive science. *Cognition*, 203, 104365.
- Palminteri, S., & Pistilli, G. (2025). The Cognitive Biases Behind Inflationary and Deflationary Claims about Large Language Models.

Writing with exposure

1. What this exercise is about

- Learn the difference between **experimental proof** and **conceptual argument**.
- Practice **thinking in public**, taking **intellectual responsibility**
- Success is convincing, and/or sparking reflection or debate. Consensus may not emerge; impact may not be directly visible; argument may end up unnoticed

2. Intellectual exposure

- Opinion writing involves **risk with accountability**
- The risk is **intellectual**: owning a position with the willingness to let the reasoning be examined, questioned, and possibly refuted
- The risk is forcing you to **examine your own assumptions** with higher accuracy
- Recklessness, outrage, controversy are high-risk, may have low-reward

3. Voice with evidence

- Use a **distinct but proportionate voice**
- Be **personal, clear, and judicious** in use of evidence
- **Avoid** rhetorical **excess**
- **Avoid** turning critique into **conflict** (personal **attacks/defenses**)

4. Well-proportioned

- You don't need to be definitive or right
- Be **clear about the strength of the claims (degrees of certainty)**, when you propose, assert, or speculate
- Tentativeness must be acknowledged, but not vague

5. Broader aim

- Write something you **endorse and understand**
- Engage deeply with an idea that matters to you
- Balance **personal insight** with **literature awareness**
- Remember **to convince or to start a discussion**. End with a clear message for your audience